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NETC-240
Lab 5 – 2-12-2025

Document each step of the lab. Take the appropriate screenshots and annotate them.

Background: In order to see meaningful changes in CPU utilization, we need to run some tasks that will waste CPU cycles. We will use the following scripts:

Bigwaste.sh	this just wastes a lot of CPU
Show.sh	this script just counts, but it also includes a time stamp for each value. This will allow some quantitative analysis and is a useful visualization

Depending on the configuration you use (CPU cores), you will need to run multiple instances of BigWaste to use up most of the CPU time. Then when you run Show, you will see it count (fast). Then you can change the priority of Show to slow down the rate that it counts.

Part 1.

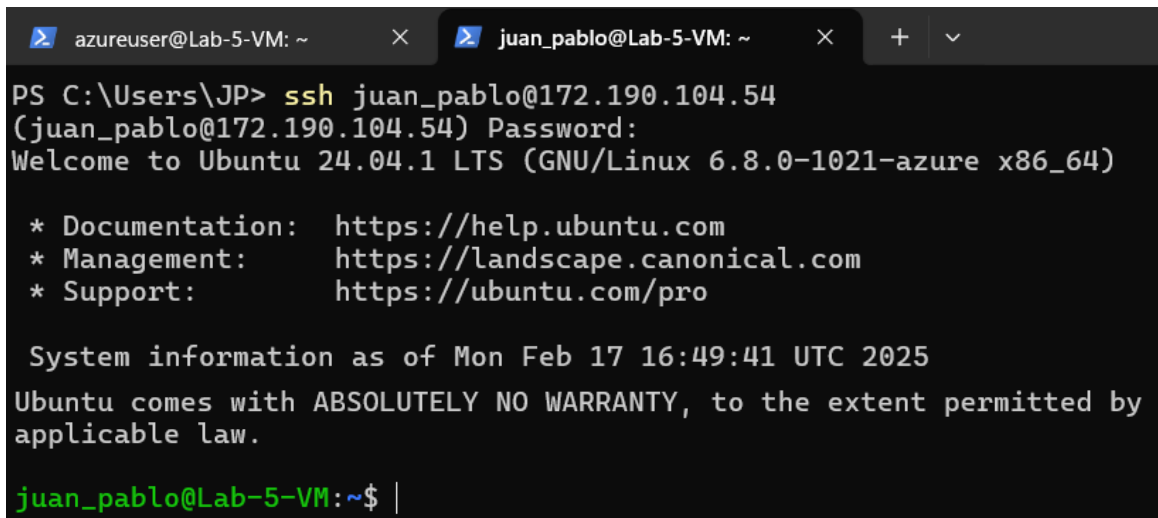
1. Create a Ubuntu 20.04 LTS VM. Create a new user using your first name as the name of the user. Use a 2nd SSH session to login using your first name account. Use the smallest size (one vCPU and smallest amount of memory)

The screenshot displays the Azure portal interface for a completed deployment. At the top, the deployment name is 'CreateVm-canonical.ubuntu-24_04-lts-server-20250217112508'. Below this, a search bar and action buttons (Delete, Cancel, Redeploy, Download, Refresh) are visible. The left sidebar shows navigation options: Overview (selected), Inputs, Outputs, and Template. The main content area features a green checkmark icon and the text 'Your deployment is complete'. Below this, deployment details are listed: 'Deployment name: CreateVm-ca...', 'Subscription: Azure for Students', and 'Resource group: Labs'. The 'Start time' is '2/17/2025, 1...'. A 'Correlation ID' of '1fc9da9e-3' is also shown. Under 'Deployment details', there are sections for 'Next steps' with links: 'Setup auto-shutdown' (Recommended), 'Monitor VM health, performance and network dependencies' (Recommended), and 'Run a script inside the virtual machine' (Recommended). At the bottom, there are two buttons: 'Go to resource' and 'Create another VM'.

Spun up an Ubuntu Server VM (above), and created the juan_pablo user after changing the ssh_config file (below):

```
azureuser@Lab-5-VM:~$ sudo nano /etc/ssh/sshd_config
azureuser@Lab-5-VM:~$ sudo adduser juan_pablo
info: Adding user `juan_pablo' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `juan_pablo' (1001) ...
info: Adding new user `juan_pablo' (1001) with group `juan_pablo (1001)' .
warn: The home directory `/home/juan_pablo' already exists. Not touching
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for juan_pablo
Enter the new value, or press ENTER for the default
    Full Name []:
    Room Number []:
    Work Phone []:
    Home Phone []:
    Other []:
Is the information correct? [Y/n]
info: Adding new user `juan_pablo' to supplemental / extra groups `users'
info: Adding user `juan_pablo' to group `users' ...
azureuser@Lab-5-VM:~$ sudo systemctl restart ssh
```

Successfully logged into name user through ssh below:



```
PS C:\Users\JP> ssh juan_pablo@172.190.104.54
(juan_pablo@172.190.104.54) Password:
Welcome to Ubuntu 24.04.1 LTS (GNU/Linux 6.8.0-1021-azure x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Mon Feb 17 16:49:41 UTC 2025
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

juan_pablo@Lab-5-VM:~$ |
```

2. Launch PowerShell 3 times and ssh into your VM on each one. Arrange the windows to show all of them simultaneously. I'll refer to these as pane1, pane2 and pane3

The image displays three terminal windows from a desktop environment. The top-left window shows system statistics (Usage of /: 5.5% of 28.02GB, Memory usage: 7%, Swap usage: 0%), user login status (Users logged in: 2, IPv4 address for eth0: 10.0.0.4), and update information (Expanded Security Maintenance for Applications is not enabled, 0 updates can be applied immediately). The top-right window shows a Windows PowerShell session with a user logging in via SSH to a Ubuntu VM. The bottom window shows similar system statistics and update information, with a slightly different timestamp for the last login (Mon Feb 17 16:53:53 2025).

```
juan_pablo@Lab-5-VM: ~
Usage of /: 5.5% of 28.02GB
Memory usage: 7%
Swap usage: 0%

Users logged in: 2
IPv4 address for eth0: 10.0.0.4

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

Last login: Mon Feb 17 16:53:46 2025 from 174.97.1.137
juan_pablo@Lab-5-VM:~$

juan_pablo@Lab-5-VM: ~
System load: 0.0
Usage of /: 5.5% of 28.02GB
Memory usage: 7%
Swap usage: 0%

Processes: 131
Users logged in: 2
IPv4 address for eth0: 10.0.0.4

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

Last login: Mon Feb 17 16:53:53 2025 from 174.97.1.137
juan_pablo@Lab-5-VM:~$

Windows PowerShell
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Install the latest PowerShell for new features and bug fixes.

PS C:\Users\JP> ssh juan_pablo@172.190.104.54
(juan_pablo@172.190.104.54) Password:
Welcome to Ubuntu 24.04.1 LTS

 * Documentation:  https://ubuntu.com/docs
 * Management:    https://ubuntu.com/manage
 * Support:       https://ubuntu.com/support

System information as of Mon Feb 17 16:53:46 2025

System load: 0.0
Usage of /: 5.5% of 28.02GB
Memory usage: 7%
Swap usage: 0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

Last login: Mon Feb 17 16:53:46 2025 from 174.97.1.137
juan_pablo@Lab-5-VM:~$
```

Panes 1, 2, and 3 simultaneously logged into the first name user.

3. In pane1 use the top command. This will show a continuously updated process list and CPU utilization.

```

> juan_pablo@Lab-5-VM: ~
top - 17:08:07 up 37 min,  5 users,  load average: 0.15, 0.07, 0.02
Tasks: 128 total,   1 running, 127 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,   0.0 sy,   0.0 ni,100.0 id,   0.0 wa,   0.0 hi,   0.0
MiB Mem :  3912.0 total,  3335.5 free,   487.9 used,   320.1 buf
MiB Swap:   0.0 total,   0.0 free,   0.0 used.  3424.1 ava

  PID USER      PR  NI   VIRT   RES    SHR S  %CPU  %MEM
   153 root        20   0     0      0      0 I    0.3   0.0
  1507 root        20   0 337356 34876 12800 S    0.3   0.9
    1 root        20   0  22752 13880  9528 S    0.0   0.3
    2 root        20   0     0      0      0 S    0.0   0.0
    3 root        20   0     0      0      0 S    0.0   0.0
    4 root         0 -20     0      0      0 I    0.0   0.0
    5 root         0 -20     0      0      0 I    0.0   0.0
    6 root         0 -20     0      0      0 I    0.0   0.0
    7 root         0 -20     0      0      0 I    0.0   0.0
    8 root        20   0     0      0      0 I    0.0   0.0
    9 root         0 -20     0      0      0 I    0.0   0.0

```

The top command in pane 1

- 4. Start Show.sh. Use the command `./show.sh | tee count.txt` in pane2. You will see the output quickly updating. It shows the count, followed by a time stamp. The display should be scrolling very quickly. This will also send the output to the count.txt file.
 - a. Stop the Show program and measure how long it took to count 10,000 times. Show your calculation and the number of counts/sec. Use 5 significant digits for the calculations and 4 significant digits for the result. Note: math mistakes reduce the lab grade by 10%.

```

> juan_pablo@Lab-5-VM: ~
top - 17:12:56 up 42 min, 5 users, load average: 0.38, 0.12,
Tasks: 127 total, 1 running, 126 sleeping, 0 stopped, 0 :
%Cpu(s): 0.3 us, 0.3 sy, 0.0 ni, 99.3 id, 0.0 wa, 0.0 hi,
MiB Mem : 3912.0 total, 3336.2 free, 487.1 used, 320
MiB Swap: 0.0 total, 0.0 free, 0.0 used. 3424

  PID USER      PR  NI  VIRT  RES  SHR S %CPU %MEM
 1092 _chrony   20   0 11192 3524 2944 S  0.3  0.1
 2719 juan_pa+  20   0 14972 6952 5120 S  0.3  0.2
 2748 juan_pa+  20   0 12356 5888 3712 R  0.3  0.1
   1 root      20   0 22752 13880 9528 S  0.0  0.3
   2 root      20   0     0     0     0 S  0.0  0.0
   3 root      20   0     0     0     0 S  0.0  0.0
   4 root       0 -20     0     0     0 I  0.0  0.0
   5 root       0 -20     0     0     0 I  0.0  0.0
   6 root       0 -20     0     0     0 I  0.0  0.0
   7 root       0 -20     0     0     0 I  0.0  0.0
   8 root      20   0     0     0     0 I  0.0  0.0

> juan_pablo@Lab-5-VM: ~
10650 17:12:52.428583057
10651 17:12:52.430817114
10652 17:12:52.433034471
10653 17:12:52.435352430
10654 17:12:52.437609088
10655 17:12:52.439853046
10656 17:12:52.442076103
10657 17:12:52.444393463
10658 17:12:52.446632220
10659 17:12:52.448843077
10660 17:12:52.451174037
10661 17:12:52.453430195
10662 17:12:52.455672552
10663 17:12:52.457896209
10664 17:12:52.460245670
10665 17:12:52.462545229
10666 17:12:52.464806187
^C
juan_pablo@Lab-5-VM: ~$ |
```

Pane 1 with the top command running, pane 2 with the show.sh program running past 10,000 times.

```
juan_pablo@Lab-5-VM: ~  
10663 17:12:52.457896209  
10664 17:12:52.460245670  
10665 17:12:52.462545229  
10666 17:12:52.464806187  
^C  
juan_pablo@Lab-5-VM:~$ head count.txt  
1 17:12:28.413057300  
2 17:12:28.415592558  
3 17:12:28.417866710  
4 17:12:28.420080560  
5 17:12:28.422393213  
6 17:12:28.424606364  
7 17:12:28.426849115  
8 17:12:28.429144067  
9 17:12:28.431341917  
10 17:12:28.433576768  
juan_pablo@Lab-5-VM:~$ grep '10000' count.txt  
10000 17:12:50.962520411  
juan_pablo@Lab-5-VM:~$ |
```

By using head, we can see that the first count happened at 17:12:28.413057300, and by using grep to find 10000, we can see that the 10,000th count happened at 17:12:50.962520411. To 5 significant digits, this is 50.963 and 28.413 seconds respectively.

Since the process took less than a minute, we can subtract the seconds and the decimals to find out how long the process took to count 10,000 times:

$50.963 - 28.413 = 22.550$. To 4 significant digits, this means it took **22.55 seconds** for the program to iterate 10,000 times.

If we take 10,000 and divide it by 22.550, we get our counts/second:

$10,000 / 22.550 = 443.46$. To 4 significant digits, this means that the program ran **443.5 times per second**.

5. Start the `bigwaste.sh` & (the & will cause the process to be run in the background). Note how much CPU time it takes. Launch additional copies of `bigwaste`, until the CPU time used is greater than 95%. (for me it was about 20 times).

```

top - 17:30:16 up 59 min, 5 users, load average: 0.49, 0.13, 0.04
Tasks: 130 total, 2 running, 128 sleeping, 0 stopped, 0 zombie
%Cpu(s): 50.2 us, 0.2 sy, 0.0 ni, 49.5 id, 0.0 wa, 0.0 hi, 0.2
MiB Mem : 3912.0 total, 3357.5 free, 465.3 used, 320.7 buf
MiB Swap: 0.0 total, 0.0 free, 0.0 used, 3446.7 ava

  PID USER      PR  NI   VIRT    RES    SHR  S  %CPU  %MEM
45699 juan_pa+  20   0   7740    3456   3200  R 100.0   0.1
1507  root      20   0 337356  34876 12800  S   0.3   0.9
2719 juan_pa+  20   0 149772  6952   5120  S   0.3   0.2
  1  root      20   0  22752  13880  9528  S   0.0   0.3
  2  root      20   0      0      0      0  S   0.0   0.0
  3  root      20   0      0      0      0  S   0.0   0.0
  4  root      0 -20      0      0      0  I   0.0   0.0
  5  root      0 -20      0      0      0  I   0.0   0.0
  6  root      0 -20      0      0      0  I   0.0   0.0
  7  root      0 -20      0      0      0  I   0.0   0.0
  8  root      20   0      0      0      0  I   0.0   0.0

  PID USER      PR  NI   VIRT    RES    SHR  S  %CPU  %MEM
894 17:30:18.852026912
895 17:30:18.854441770
896 17:30:18.856814626
897 17:30:18.859187682
898 17:30:18.861533637
899 17:30:18.863931294
900 17:30:18.866303850
901 17:30:18.868680106
902 17:30:18.871015762
903 17:30:18.873358517
904 17:30:18.875709073
905 17:30:18.878091429
906 17:30:18.880464085
907 17:30:18.882803040
908 17:30:18.885157396
909 17:30:18.887551453
910 17:30:18.889903408
911 17:30:18.892255664
912 17:30:18.894765023

Windows PowerShell
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Install the latest PowerShell for new features and improvements!
https://aka.ms/PowerShellLatest

PS C:\Users\JP> ssh juan_pablo@172.190.104.54
(juan_pablo@172.190.104.54) Password:
Welcome to Ubuntu 24.04.1 LTS (GNU/Linux 6.8.0-26-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/support

System information as of Mon Feb 17 16:5
System load:  0.0          Process
Usage of /:   5.5% of 28.02GB Users logged in: 1
Memory usage: 7%          IPv4 address: 172.190.104.54
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo apt install --install-recommends ubuntu-esm-apps

The list of available updates is more than 1000.
To check for new updates run: sudo apt update

Last login: Mon Feb 17 16:49:42 2025 from 172.190.104.54
juan_pablo@Lab-5-VM:~$ nano bigwaste.sh
juan_pablo@Lab-5-VM:~$ chmod 777 bigwaste.sh
juan_pablo@Lab-5-VM:~$ ./bigwaste.sh &
[1] 45699
juan_pablo@Lab-5-VM:~$
```

With the smaller machine that I spun, it only took one copy of `bigwaste` to get 100% of the CPU.

6. At this point, `pane1` is running `top`, `pane2` is showing the `show.sh` output. Using `pane3`, `renice` the `show.sh` process with the following priorities (20, 10, 0, -10, -19). For each trial calculate the counts/second (as in step 4). Use the time difference between at least 1000 counts. Show the results in table format.

```

juan_pablo@Lab-5-VM:~$ sed -n '164000,165000p' count.txt > prio19.txt
juan_pablo@Lab-5-VM:~$ nano prio19.txt
juan_pablo@Lab-5-VM:~$ head prio19.txt
164000 17:36:40.871347407
164001 17:36:40.873753461
164002 17:36:40.876153715
164003 17:36:40.878499567
164004 17:36:40.880923121
164005 17:36:40.883233072
164006 17:36:40.885562924
164007 17:36:40.887895876
164008 17:36:40.890287930
164009 17:36:40.892708584
juan_pablo@Lab-5-VM:~$ tail prio19.txt
164991 17:36:43.216458679
164992 17:36:43.218828632
164993 17:36:43.221183784
164994 17:36:43.223528636
164995 17:36:43.225902289
164996 17:36:43.228265842
164997 17:36:43.230660395
164998 17:36:43.233051049
164999 17:36:43.235320799
165000 17:36:43.237649451
juan_pablo@Lab-5-VM:~$ sed -n '223000,224000p' count.txt > prio10.txt
juan_pablo@Lab-5-VM:~$ sed -n '237000,238000p' count.txt > prio0.txt
juan_pablo@Lab-5-VM:~$ sed -n '259000,260000p' count.txt > prion10.txt
juan_pablo@Lab-5-VM:~$ sed -n '270000,271000p' count.txt > prion20.txt

```

Above is the process by which I gathered the data per for the count script – I would renice the process and pick a range of 1000 to track and wrote it down.

For priority 20, the range was 164000-165000,
 For priority 10, the range was 223000-224000,
 For priority 0, the range was 237000-238000,
 For priority -10, the range was 259000-260000,
 For priority -20, the range was 270000-271000.

I stored the values into a separate text file so I could use the head and tail method to get the first and last lines in the file and pull the data I needed for the calculations from there. The screenshot of this method is below:


```

juan_pablo@Lab-5-VM:~$ head -n 1 prio19.txt
164000 17:36:40.871347407
juan_pablo@Lab-5-VM:~$ tail -n 1 prio19.txt
165000 17:36:43.237649451
juan_pablo@Lab-5-VM:~$ head -n 1 prio10.txt
223000 17:39:00.664528267
juan_pablo@Lab-5-VM:~$ tail -n 1 prio10.txt
224000 17:39:03.037249986
juan_pablo@Lab-5-VM:~$ head -n 1 prio0.txt
237000 17:39:33.774771882
juan_pablo@Lab-5-VM:~$ tail -n 1 prio0.txt
238000 17:39:36.128889074
juan_pablo@Lab-5-VM:~$ head -n 1 prion10.txt
259000 17:40:26.825662291
juan_pablo@Lab-5-VM:~$ tail -n 1 prion10.txt
260000 17:40:29.183622396
juan_pablo@Lab-5-VM:~$ head -n 1 prion20.txt
270000 17:40:52.607747398
juan_pablo@Lab-5-VM:~$ tail -n 1 prion20.txt
271000 17:40:54.963300314

```

The data is summarized in the table below:

Priority	Start time (sec)	End Time (sec)	Time per 1000	Counts/sec
20	40.871	43.238	2.367 seconds	422.5
10	0.6645	3.0372	2.373 seconds	421.5
0	33.775	36.129	2.354 seconds	424.8
-10	26.826	29.184	2.358 seconds	424.1
-19	52.608	54.963	2.355	424.6

It's interesting that the counts/sec generally show an increase as the priority gets higher, though a couple were faster or slower than expected. Still, very consistent with the concept of the prioritization.

7. Kill all of the processes.

```
top - 20:36:01 up 4:05, 1 user, load average: 0.00, 0.00, 0.00
Tasks: 113 total, 1 running, 112 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni, 100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 3912.0 total, 3060.4 free, 485.1 used, 602.2 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used, 3426.9 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1507	root	20	0	338380	35004	12800	S	0.3	0.9	0:10.17	python3
1	root	20	0	22692	14008	9528	S	0.0	0.3	0:03.99	systemd
2	root	20	0	0	0	0	S	0.0	0.0	0:00.00	kthreadd
3	root	20	0	0	0	0	S	0.0	0.0	0:00.00	pool_workq+
4	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R+
5	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R+

The processes have been killed.

Part 2.

1. Create a Cron job that will run every 1 minute. The job should ping an IP address or name, save the numeric result with a time stamp to a log file in /var/log. Pick a site that returns at least 40 mS of latency. Show the following:
 - a. job in crontab.
 - b. Partial of the log file, showing 10 seconds of output

```
azureuser@Lab-5-VM:~$ cat /etc/cron.d/pinger.sh
#!/bin/bash

# IP to ping
IP="8.8.8.8"

# Number of pings
COUNT=3

# Log file
LOGFILE="/var/log/pinger.log"

# Perform pings
ping -c $COUNT $IP > ping_output.txt

# Grep average RTT
average=$(grep "rtt" ping_output.txt | awk '{print $4}' | cut -d '/' -f 2)

# Remove units
ms=$(echo $average | cut -d '.' -f 1)

# Timestamp
TIMESTAMP=$(date "+%Y-%m-%d %H:%M:%S")

# Write ping time and timestamp to log
echo "$ms $TIMESTAMP" >> $LOGFILE

# Clean up
rm ping_output.txt
```

I added the pinger.sh script to the /etc/cron.d folder (above) and added execute permissions to it. (below)

```
azureuser@Lab-5-VM:~$ sudo chmod +x /etc/cron.d/pinger.sh
azureuser@Lab-5-VM:~$ crontab -e
no crontab for azureuser - using an empty one
crontab: installing new crontab
azureuser@Lab-5-VM:~$ |
```

I then opened up crontab -e and added the path to my executable as well as prefixed it with * * * * *, which will make it run every minute. (below)

```
GNU nano 7.2 /tmp/crontab.dLd35S/crontab
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h dom mon dow  command
* * * * * /etc/cron.d/pinger.sh >> /var/log/pinger.log 2>&1
```

Below you can see that the job is running via CRON:

```
Feb 17 21:36:01 Lab-5-VM CRON[648349]: (azureuser) CMD (/etc/cron.d/pinger.sh >> /var/log/pinger.log 2>&1)
Feb 17 21:36:01 Lab-5-VM CRON[648347]: (CRON) info (No MTA installed, discarding output)
```

Here is part of the log file:

```
azureuser@Lab-5-VM:/var/log$ head pinger.log
1 2025-02-17 21:41:03
1 2025-02-17 21:42:03
1 2025-02-17 21:43:03
1 2025-02-17 21:44:03
1 2025-02-17 21:45:03
1 2025-02-17 21:46:03
```

Part 3.

1. Use the tcpdump command to capture packets on the VM NIC

```
22:13:52.066726 IP lab-5-vm.internal.cloudapp.net.ssh > nw1-dsl-74-215-81-41.fuse.net.49328: Flags [P.], seq 1680833:1681165, ack 4176, win 1018, length 332
22:13:52.154518 IP nw1-dsl-74-215-81-41.fuse.net.49328 > lab-5-vm.internal.cloudapp.net.ssh: Flags [.] , ack 1681165, win 510, length 0
22:13:52.170738 IP lab-5-vm.internal.cloudapp.net.ssh > nw1-dsl-74-215-81-41.fuse.net.49328: Flags [D.], seq 1681165:1681497, ack 4176, win 1018, length 332
22:13:52.215598 IP lab-5-vm.internal.cloudapp.net.39480 > 168.63.129.16.http: Flags [S], seq 3968104789, win 64240, options [mss 1460,sackOK,TS val 38432131
94 ecr 0,nop,wscale 7], length 0
22:13:52.216377 IP 168.63.129.16.http > lab-5-vm.internal.cloudapp.net.39480: Flags [S.], seq 797205526, ack 3960104790, win 65535, options [mss 1460,nop,ws
cale 8,sackOK,TS val 4011538522 ecr 3843213194], length 0
22:13:52.216401 IP lab-5-vm.internal.cloudapp.net.39480 > 168.63.129.16.http: Flags [.] , ack 1, win 502, options [nop,nop,TS val 3843213195 ecr 4011538522],
length 0
22:13:52.216436 IP lab-5-vm.internal.cloudapp.net.39480 > 168.63.129.16.http: Flags [P.], seq 1:201, ack 1, win 502, options [nop,nop,TS val 3843213195 ecr
4011538522], length 200: HTTP: GET /machine/?comp=goalstate HTTP/1.1
22:13:52.219921 IP 168.63.129.16.http > lab-5-vm.internal.cloudapp.net.39480: Flags [FP.], seq 1:2270, ack 201, win 16387, options [nop,nop,TS val 401153852
5 ecr 3843213195], length 2269: HTTP: HTTP/1.1 200 OK
22:13:52.219955 IP lab-5-vm.internal.cloudapp.net.39480 > 168.63.129.16.http: Flags [.] , ack 2271, win 537, options [nop,nop,TS val 3843213198 ecr 401153852
5], length 0
22:13:52.220196 IP lab-5-vm.internal.cloudapp.net.39480 > 168.63.129.16.http: Flags [F.], seq 201, ack 2271, win 537, options [nop,nop,TS val 3843213198 ecr
4011538525], length 0
22:13:52.220687 IP 168.63.129.16.http > lab-5-vm.internal.cloudapp.net.39480: Flags [.] , ack 202, win 16387, options [nop,nop,TS val 4011538527 ecr 38432131
98], length 0
22:13:52.220795 IP lab-5-vm.internal.cloudapp.net.41854 > 168.63.129.16.32526: Flags [S], seq 4078098395, win 64240, options [mss 1460,sackOK,TS val 3843213
199 ecr 0,nop,wscale 7], length 0
22:13:52.221380 IP 168.63.129.16.32526 > lab-5-vm.internal.cloudapp.net.41854: Flags [S.], seq 2291156848, ack 4078098396, win 65535, options [mss 1460,nop,
wscale 8,sackOK,TS val 4011538527 ecr 3843213199], length 0
22:13:52.221395 IP lab-5-vm.internal.cloudapp.net.41854 > 168.63.129.16.32526: Flags [.] , ack 1, win 502, options [nop,nop,TS val 3843213200 ecr 4011538527],
length 0
22:13:52.221426 IP lab-5-vm.internal.cloudapp.net.41854 > 168.63.129.16.32526: Flags [P.], seq 1:438, ack 1, win 502, options [nop,nop,TS val 3843213200 ecr
4011538527], length 437
22:13:52.225266 IP 168.63.129.16.32526 > lab-5-vm.internal.cloudapp.net.41854: Flags [FP.], seq 1:146, ack 438, win 16386, options [nop,nop,TS val 401153853
1 ecr 3843213200], length 145
22:13:52.225446 IP lab-5-vm.internal.cloudapp.net.41854 > 168.63.129.16.32526: Flags [F.], seq 438, ack 147, win 501, options [nop,nop,TS val 3843213204 ecr
4011538531], length 0
22:13:52.225720 IP 168.63.129.16.32526 > lab-5-vm.internal.cloudapp.net.41854: Flags [.] , ack 439, win 16386, options [nop,nop,TS val 4011538531 ecr 3843213
204], length 0
```

Then I ran tcpdump as follows:

```
azureuser@Lab-5-VM:~$ sudo tcpdump -i eth0 -w /home/juan_pablo/tcpdumpfile.pcap
tcpdump: listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
^C425 packets captured
430 packets received by filter
0 packets dropped by kernel
azureuser@Lab-5-VM:~$ sudo chmod 777 /home/juan_pablo/tcpdumpfile.pcap
```

The output from tcpdump as it's running (above)

2. Use Nmap to scan the public IP address from your home.

- Perform an operating system detection
- Scan for open ports

```
PS C:\program files (x86)\Nmap> .\nmap -v -O -Pn 172.190.104.54
Host discovery disabled (-Pn). All addresses will be marked 'up' and scan times may be slower.
Starting Nmap 7.95 ( https://nmap.org ) at 2025-02-17 17:14 Eastern Standard Time
Initiating Parallel DNS resolution of 1 host. at 17:14
Completed Parallel DNS resolution of 1 host. at 17:14, 0.18s elapsed
Initiating SYN Stealth Scan at 17:14
Scanning 172.190.104.54 [1000 ports]
Discovered open port 53/tcp on 172.190.104.54
Discovered open port 22/tcp on 172.190.104.54
Completed SYN Stealth Scan at 17:14, 5.20s elapsed (1000 total ports)
Initiating OS detection (try #1) against 172.190.104.54
Retrying OS detection (try #2) against 172.190.104.54
Nmap scan report for 172.190.104.54
Host is up (0.030s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT      STATE SERVICE
22/tcp    open  ssh
53/tcp    open  domain
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: general purpose/router
Running (JUST GUESSING): Linux 4.X|5.X (92%), MikroTik RouterOS 7.X (85%)
OS CPE: cpe:/o:linux:linux_kernel:4 cpe:/o:linux:linux_kernel:5 cpe:/o:mikrotik:routeros:7 cpe:/o:linux:linux_kernel:5.6.3
Aggressive OS guesses: Linux 4.19 - 5.15 (92%), Linux 4.15 - 5.19 (85%), Linux 5.0 - 5.14 (85%), MikroTik RouterOS 7.2 - 7.5 (Linux 5.6.3) (85%), Linux 4.15
(85%)
No exact OS matches for host (test conditions non-ideal).
Uptime guess: 23.896 days (since Fri Jan 24 19:43:58 2025)
TCP Sequence Prediction: Difficulty=254 (Good luck!)
IP ID Sequence Generation: All zeros

Read data files from: C:\program files (x86)\Nmap
OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 10.35 seconds
Raw packets sent: 2087 (96.964KB) | Rcvd: 53 (2.832KB)
```

Running nmap from my Windows PC (above), with the tags -O for operating system, -v for verbose, and -Pn because I know the host is up, so it will proceed with port scanning no matter what.

3. After 20 seconds, stop the capture

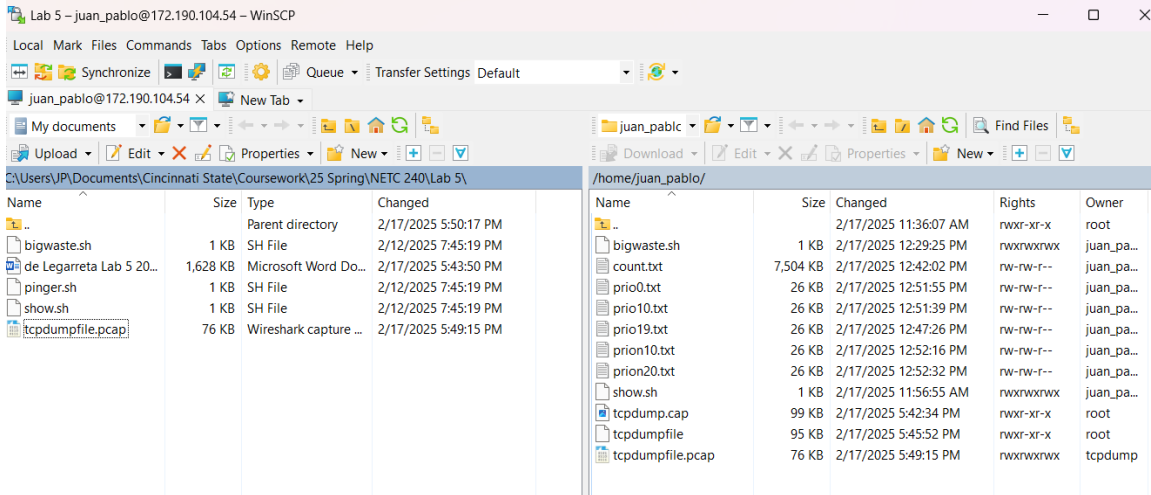
4. Transfer the file to your home PC and open it with Wireshark

```

azureuser@Lab-5-VM:~$ sudo tcpdump -i eth0 -w /home/juan_pablo/tcpdumpfile.pcap
tcpdump: listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
^C425 packets captured
430 packets received by filter
0 packets dropped by kernel
azureuser@Lab-5-VM:~$ sudo chmod 777 /home/juan_pablo/tcpdumpfile.pcap

```

I saved the file tcpdumpfile.pcap to the juan_pablo user and changed its permissions, then accessed it with WinSCP (below):



5. Find a TCP packets, where the Flags are set to - Acknowledgment set

No.	Time	Source	Destination	Protocol	Length	Info
78	0.996626	10.0.0.4	168.63.129.16	TCP	66	40976 → 80 [ACK] Seq=201 Ack=2271 Win=68736 Len=0 TSval=3845308016 TSecr=4013633295
79	0.996642	10.0.0.4	74.215.81.41	SSH	90	Server: Encrypted packet (len=36)
80	0.996865	10.0.0.4	168.63.129.16	TCP	66	40976 → 80 [FIN, ACK] Seq=201 Ack=2271 Win=68736 Len=0 TSval=3845308016 TSecr=4013633295
81	0.997136	168.63.129.16	10.0.0.4	TCP	66	80 → 40976 [ACK] Seq=2271 Ack=202 Win=4195072 Len=0 TSval=4013633296 TSecr=3845308016
83	0.998168	168.63.129.16	10.0.0.4	TCP	74	32526 → 37476 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM TSval=3845308016 TSecr=4013633297
84	0.998187	10.0.0.4	168.63.129.16	TCP	66	37476 → 32526 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=3845308018 TSecr=4013633297
85	0.998217	10.0.0.4	168.63.129.16	HTTP	503	GET /vmSettings HTTP/1.1
86	1.002696	168.63.129.16	10.0.0.4	HTTP	211	HTTP/1.1 304 Not Modified
87	1.002776	10.0.0.4	168.63.129.16	TCP	66	37476 → 32526 [FIN, ACK] Seq=438 Ack=147 Win=64128 Len=0 TSval=3845308022 TSecr=4013633300
88	1.003144	168.63.129.16	10.0.0.4	TCP	66	32526 → 37476 [ACK] Seq=147 Ack=439 Win=4194816 Len=0 TSval=4013633302 TSecr=3845308022
90	1.003549	168.63.129.16	10.0.0.4	TCP	74	32526 → 37476 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM TSval=4013633303 TSecr=3845308022

You can see above that packet 84 has an ACK flag. Some are [SYN,ACK], some are [FIN, ACK].

- Find one other example of a different TCP flag setting that is used

No.	Time	Source	Destination	Protocol	Length	Info
72	0.993636	10.0.0.4	168.63.129.16	TCP	74	40976 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=3845308013 TSecr=0 WS=128
73	0.994405	168.63.129.16	10.0.0.4	TCP	74	80 → 40976 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM TSval=4013633293 TSecr=3845308013
82	0.997807	10.0.0.4	168.63.129.16	TCP	74	37476 → 32526 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=3845308017 TSecr=0 WS=128
83	0.998168	168.63.129.16	10.0.0.4	TCP	74	32526 → 37476 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM TSval=4013633297 TSecr=3845308017
89	1.003162	10.0.0.4	168.63.129.16	TCP	74	37476 → 32526 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=3845308022 TSecr=0 WS=128
90	1.003549	168.63.129.16	10.0.0.4	TCP	74	32526 → 37476 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM TSval=4013633303 TSecr=3845308022
96	1.004647	10.0.0.4	168.63.129.16	TCP	74	40990 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=3845308024 TSecr=0 WS=128
97	1.005017	168.63.129.16	10.0.0.4	TCP	74	80 → 40990 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM TSval=4013633304 TSecr=3845308024
104	1.008858	10.0.0.4	168.63.129.16	TCP	74	37494 → 32526 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=3845308028 TSecr=0 WS=128
106	1.009347	168.63.129.16	10.0.0.4	TCP	74	32526 → 37494 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM TSval=4013633308 TSecr=3845308028

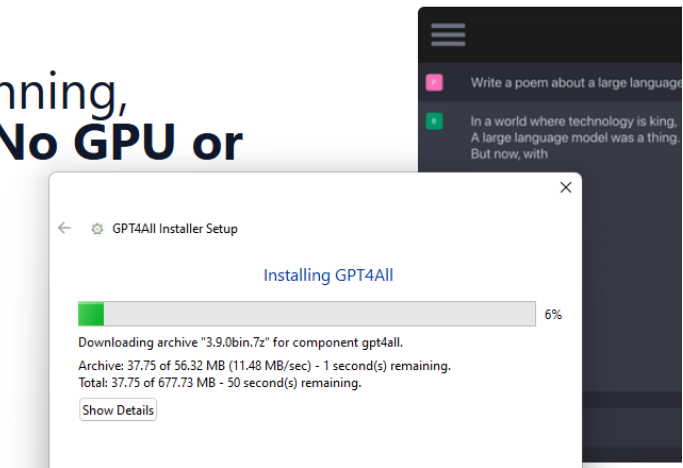
You can see above that the packetets have a SYN flag – some like 83 are [SYN, ACK], some are just [SYN] like 72.

Part 4.

1. Install GPT4all on your lab PC. Quit the program once it is installed

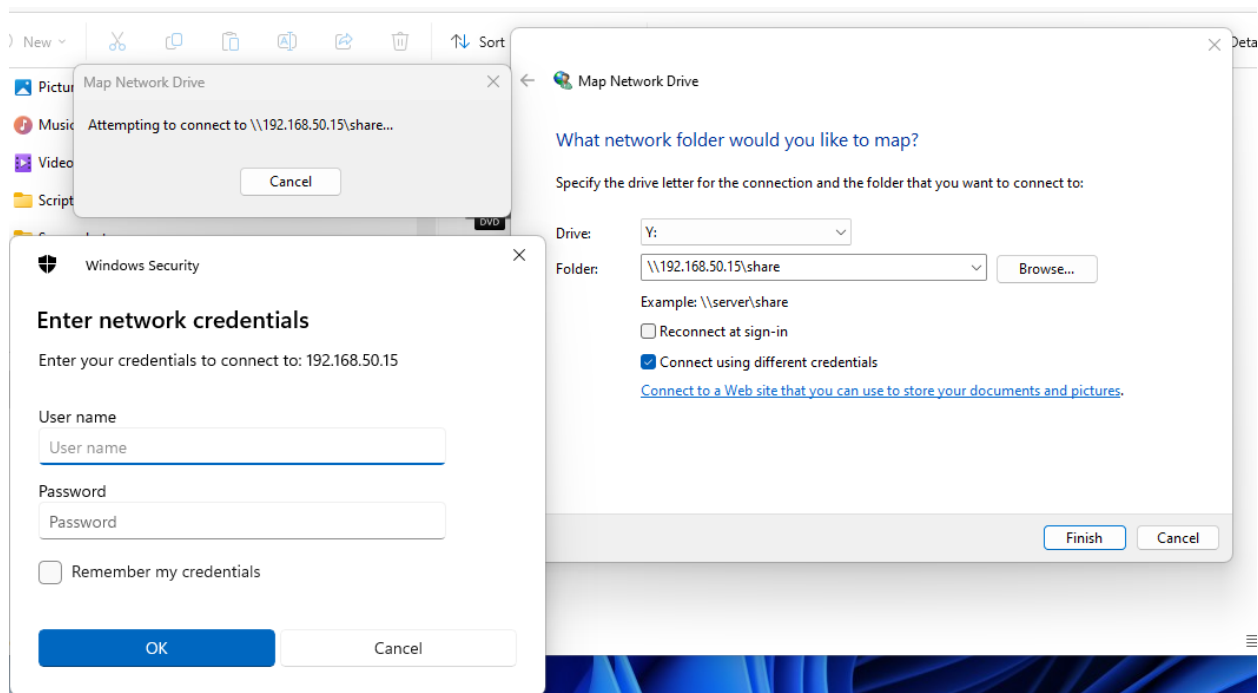
GPT4All

A free-to-use, locally running,
privacy-aware chatbot. **No GPU or
internet required.**

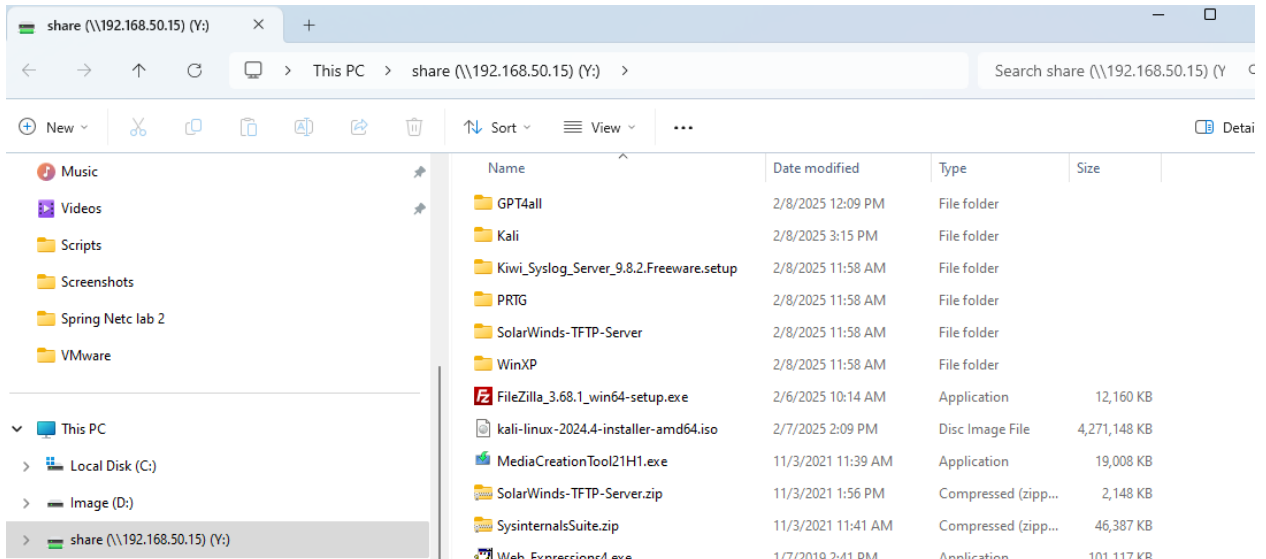


This is the installation of GPT4ALL

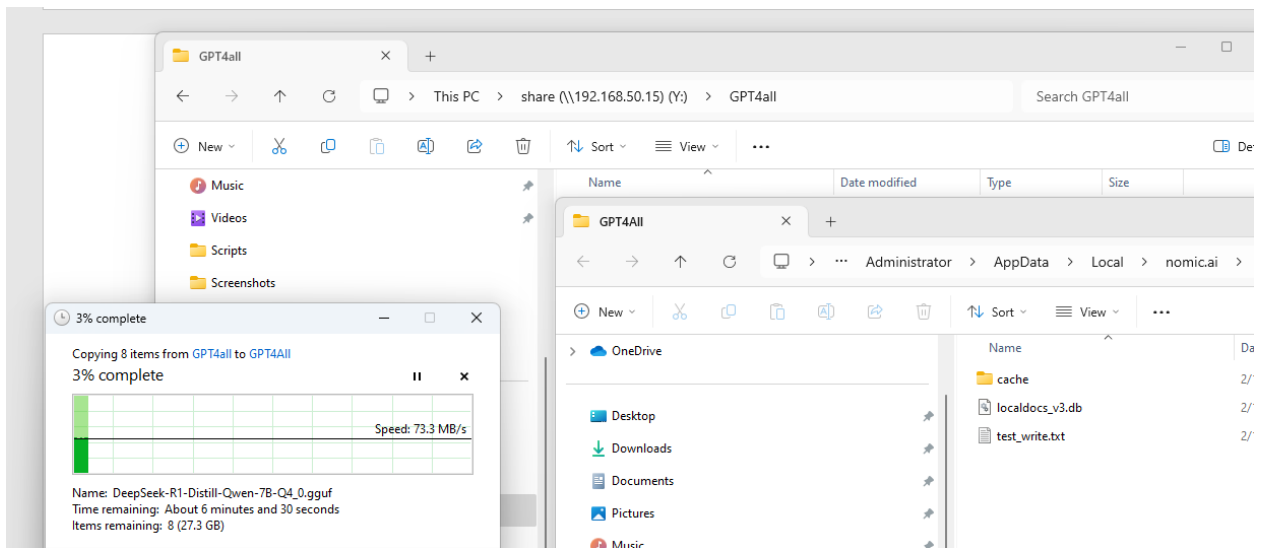
2. Map a network drive to 192.168.50.15, folder name is "share". Copy the GPT4All folder to your PC.



Above you can see the network drive being mapped, with the successful mapped drive launched below:



Below is a screenshot of copying the files directly from the network drive to the user/appdata/local/nomic.ai folder:



3. Disconnect the mapped drive
4. Copy the files in the GPT4all folder to the user, appdata, local, nomic.ai folder

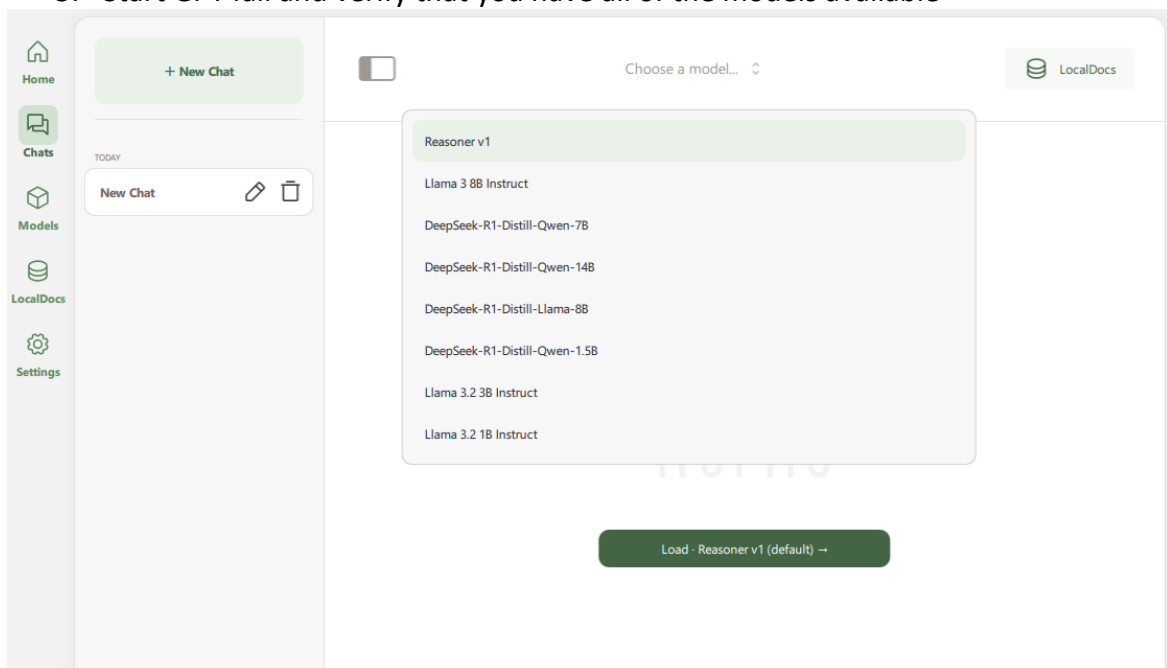
erator > AppData > Local > nomic.ai > GPT4All > Search GPT4All

Sort View ...

Name	Date modified	Type	Size
cache	2/12/2025 7:58 PM	File folder	
localdocs_v3.db	2/12/2025 7:58 PM	Data Base File	76 KB
test_write.txt	2/12/2025 7:58 PM	Text Document	0 KB
DeepSeek-R1-Distill-Qwen-7B-Q4_0.gguf	2/12/2025 8:00 PM	GGUF File	4,339,962 KB
Llama-3.2-1B-Instruct-Q4_0.gguf	2/12/2025 8:05 PM	GGUF File	754,909 KB
Llama-3.2-3B-Instruct-Q4_0.gguf	2/12/2025 8:07 PM	GGUF File	1,876,865 KB
DeepSeek-R1-Distill-Qwen-14B-Q4_0.gguf	2/12/2025 8:15 PM	GGUF File	8,344,012 KB
Meta-Llama-3-8B-Instruct.Q4_0.gguf	2/12/2025 8:16 PM	GGUF File	4,552,466 KB
qwen2.5-coder-7b-instruct-q4_0.gguf	2/12/2025 8:17 PM	GGUF File	4,327,530 KB
DeepSeek-R1-Distill-Llama-8B-Q4_0.gguf	2/12/2025 8:18 PM	GGUF File	4,566,303 KB
DeepSeek-R1-Distill-Qwen-1.5B-Q4_0.gguf	2/12/2025 8:18 PM	GGUF File	1,043,758 KB
log-prev.txt	2/12/2025 8:16 PM	Text Document	1 KB
log.txt	2/12/2025 8:28 PM	Text Document	0 KB

You can see the copied files on the finder window

5. Start GPT4all and verify that you have all of the models available



6. Use two specific models, Llama 1B Instruct and the DeepSeek 14B model

The goal of this part is to examine performance, limitations (Context Window, knowledge base, etc.) and expose other issues. This cannot be done using simple prompts.

7. Create several well constructed, thoughtful prompts that show the differences in the model outputs. Explain your findings. Consider using AI to help you with this process.

Specifically address the following:

- a. Context window
- b. Speed (consider measuring the time between when you press enter and the first results begin to flow. Use your smartphone stopwatch. Speed = number of words in the prompt / elapsed time)

 ChatGPT

help me to create a well-constructed, thoughtful prompt that shows differences in LLM model outputs

To create a well-constructed, thoughtful prompt that can highlight the differences in outputs between large language models (LLMs), you should design a prompt that encourages the model to address multiple facets of a question or topic, ideally allowing for variations in style, tone, complexity, and reasoning.

Here's a prompt example that will encourage different models to provide varied outputs:

Prompt:

Imagine you are a storyteller asked to explain the concept of "artificial intelligence" to a group of people. However, your audience includes different types of people: a child, a college student, and a seasoned engineer. Please provide three different responses, each tailored to one of these audiences. You should adjust your language, examples, and level of detail for each group, and provide an explanation that matches their understanding and interest in the topic. Afterward, describe how the same concept might be conveyed in a formal academic paper.

I asked ChatGPT to create a thoughtful prompt, seen above. It suggested a prompt to ask the AI to explain artificial intelligence as a storyteller for different audiences and use unique language, examples, and level of detail for each group. I used this prompt with the DeepSeek and Llama AI.

The Llama model took 12 seconds to start to respond.



Llama 3.2 1B Instruct



You

Imagine you are a storyteller asked to explain the concept of "artificial intelligence" to a group of people. However, your audience includes different types of people: a child, a college student, and a seasoned engineer. Please provide three different responses, each tailored to one of these audiences. You should adjust your language, examples, and level of detail for each group, and provide an explanation that matches their understanding and interest in the topic. Afterward, describe how the same concept might be conveyed in a formal academic paper.



GPT4All Llama 3.2 1B Instruct

For the Child:

Imagine you have a toy robot that can play games with you, like hide-and-seek or tag. It's really good at it too! But sometimes it makes mistakes and does things on its own without asking for permission.

Artificial intelligence is kind of like how your robot plays games - but instead of just playing the same old game over and over again, AI can learn to play new ones too!

It uses special computers that are super fast and smart. They're like super-powerful calculators that can think really quickly and make decisions on their own.

For example, imagine you have a toy robot that's programmed to pick up toys from the floor when it needs them. It might get distracted by something else while playing with its favorite game, but then remember where all the other toys are so it knows exactly what to do next!

AI is like having your very own super-smart assistant that can help you in lots of different ways.

For the College Student:

Artificial intelligence (AI) refers to a broad range of technologies and systems designed to simulate human-like intelligent behavior, often with the goal of solving complex problems or performing tasks autonomously. The term encompasses various subfields, including machine learning, deep learning, natural language processing, computer vision, robotics, and more.

Think of AI as a tool that enables machines to learn from data, reason, and interact with humans in ways that are indistinguishable from human intelligence. This is achieved through the development of sophisticated algorithms, neural networks, and statistical models that enable computers to process vast amounts of information and make decisions based on patterns and relationships.

I gave the following follow up prompts:

"for the college student, create an outline to study the story"

The outline was fine, but not super detailed or thorough. Kind of generic

"what would the title of the story be for a child?"

There were clever names for the story that followed what had been proposed earlier.

"what allegory could you make about this story?"

At this point, the AI got a little repetitive and offered essentially a slight variation on the story it had shared earlier.

"there's a lot of repetition here. Can you give me some more variety?"

It changed some of the vocabulary but the examples all amounted to the same allegory/metaphor.

"what would be the central part of the original AI story for the engineer?"

At this point, the AI based the answer on the allegory and began to lose the context. The context window's limits began to show here.

The DeepSeek14b model took 38 seconds to populate the first sentences.



imagine you are a storyteller asked to explain the concept of 'artificial intelligence' to a group of people. However, your audience includes different types of people: a child, a college student, and a seasoned engineer. Please provide three different responses, each tailored to one of these audiences. You should adjust your language, examples, and level of detail for each group, and provide an explanation that matches their understanding and interest in the topic. Afterward, describe how the same concept might be conveyed in an academic paper



Thought for 45 seconds >

Explaining Artificial Intelligence (AI) to Different Audiences

1. **To a Child:** Imagine you have a toy robot that can talk and recognize your voice! AI is like magic where machines learn from experience, just like how you learn games or new things by playing with toys. It's all about teaching computers to do tasks better over time without being told exactly what to do.
2. **To a College Student:** AI uses complex math and patterns to make predictions and decisions. Think of it as when your phone suggests songs based on the words you type or how you speak. Undergraduates can explore algorithms like machine learning, where systems improve through practice, similar to how neural networks mimic human brain connections.
3. **To a Seasoned Engineer:** AI involves creating models that learn from data without explicit programming for each task

The model also was more brief than the Llama model's. I also felt that they were more specific and creative:

Explaining Artificial Intelligence (AI) to Different Audiences

1. **To a Child:** Imagine you have a toy robot that can talk and recognize your voice! AI is like magic where machines learn from experience, just like how you learn games or new things by playing with toys. It's all about teaching computers to do tasks better over time without being told exactly what to do.
2. **To a College Student:** AI uses complex math and patterns to make predictions and decisions. Think of it as when your phone suggests songs based on the words you type or how you speak. Undergraduates can explore algorithms like machine learning, where systems improve through practice, similar to how neural networks mimic human brain connections.
3. **To a Seasoned Engineer:** AI involves creating models that learn from data without explicit programming for each task —like self-driving cars adjusting speed based on traffic patterns or facial recognition systems. Engineers delve into advanced topics such as deep learning and optimization techniques crucial for developing efficient algorithms, addressing challenges like computational complexity.

Academic Paper Structure:

An academic paper would structure AI explanation with formal sections:

- **Introduction:** Define AI technically, mentioning its evolution from machine learning to areas like deep learning.
- **Methodology/Approaches:** Detail methods such as supervised vs unsupervised learning and neural networks using equations for clarity.
- **Results & Applications:** Discuss successful implementations in healthcare or finance, highlighting algorithmic advancements with specific examples.
- **Challenges & Ethical Considerations:** Explore issues like bias through case studies on facial recognition systems used globally.
- **Conclusion:** Summarize AI's transformative impact and emphasize ongoing research to address challenges for future innovations.

When asked the follow up questions, the DeepSeek Model had the following responses:

“for the college student, create an outline to study the story”

Thought for 31 seconds, created a thorough outline, but it didn’t quite match up to the earlier answer. It felt more objective and less tied to the story that was proposed by the AI in the first prompt.

Thought for 31 seconds >

Sure! Here's a simple outline tailored for a college student to study the story of artificial intelligence:

Outline: The Story of Artificial Intelligence**1. Introduction to AI**

- What is AI? (Think of it as "intelligent" machines that can perform tasks like learning, problem-solving, and decision-making.)
- Examples in everyday life: smartphones, voice assistants, recommendation systems on platforms like Netflix or Spotify.

2. History of AI

- When did AI start? Early ideas date back to the 1950s with figures like Alan Turing (the "father of computer science") and Claude Shannon.
- Key milestones:
 - 1973: First general-purpose AI program called "Tatooine" tried to solve simple math problems but failed because it couldn't handle ambiguity in language.
 - 2015: IBM's Watson supercomputer beat world champions at Jeopardy! (This showed how advanced AI can be, even if still far from human-level intelligence.)

3. How AI Works

- Simple explanation of machine learning and algorithms without getting too technical:
 - Computers learn patterns in data to make predictions or decisions.

“what would the title of the story be for a child?”

It thought for 62 seconds and came up with a rather good title that actually pretty closely matched the context of the story proposed for the child earlier.

Thought for 62 seconds >

Title: "The Whizbit Who Knows Your Mind"

This title effectively captures the essence of artificial intelligence in an imaginative way, making it relatable to children by presenting AI as a friendly helper who understands their thoughts.

“what allegory could you make about this story?”

It thought for 32 seconds, then gave suggestions instead of coming up with a story or allegory. It's not necessarily wrong, but it just feels a little shallow.

Thought for 32 seconds >

To create an allegory based on your story involving a child's experience with technology, you could use robots as symbolic characters representing progress and human values. Here's how:

1. **Characters:** Use friendly, helpful robots (like R2D2 from Star Wars) to symbolize positive aspects of AI development.
2. **Conflict:** Introduce challenges that the robot faces, mirroring the struggles a child might experience when learning new technology.
3. **Resolution:** Show how the child overcomes these challenges with patience and understanding—symbolizing resilience in adapting to technological changes while emphasizing human values like kindness and logic.

This approach uses robots as modern symbols for progress and contrasts them with human qualities that ensure positive outcomes, making it relatable and meaningful for a child's perspective.

“what would be the central part of the original AI story for the engineer?”

It thought for 45 seconds and responded with a decent story. It actually still has a good context window.

Thought for 45 seconds >

The central part of the original AI story for an engineer could revolve around **“Optimizing Industrial Processes with AI Technology.”** This theme highlights how AI can enhance efficiency in manufacturing by automating repetitive tasks. Engineers would likely focus on the technical aspects, such as algorithm design and system integration to achieve seamless automation without human intervention.

Summary:

The central part of the original AI story for an engineer is **“Optimizing Industrial Processes with AI Technology.”** This theme emphasizes how AI can revolutionize manufacturing by automating tasks through advanced algorithms and integrated systems.

I did the same exercise with the following prompts:

1. Imagine that you need to explain quantum computing in a song for three levels of audience: a child, a college student, and a subject matter expert. Use appropriate language, terms, and concepts for each level and make sure the song can be played in 4/4 time. Afterward, describe how a research paper would analyze the song including music theory terminology and structure.

I wanted to test the AI's generative creativity and see if it could create something artistic while following a couple of rules and requests.

In the end, the AI presented various rhyming verses that loosely related to the topic.

Thought for 59 seconds >

Song Explanation:

1. For Children:

- *Verse 1:* "Imagine a tiny world where things can be in two places at once, like a top spinning so fast you see both sides! Qubits are our friends here, spinning and twirling with glee."
- *Chorus:* "Quantum computing is fun, it's the future we've been yearning for. With qubits dancing side by side, who knows what we'll discover!"

2. For College Students:

- *Verse 1:* "Picture a particle so small and strange, existing in superposition where nothing's quite right! Qubits are our allies here, building the future with might."
- *Chorus:* "Quantum computing's power is vast, it's like entanglement when you least expect. With qubits linking together, we're touching the stars!"

3. For Subject Matter Experts:

- *Verse 1:* "In this quantum symphony, we hear a call to action—explore new frontiers where traditional logic can't go! Qubits and entanglement guide us through the maze."
- *Chorus:* "Quantum computing's potential is boundless, it's like a wave in the ocean of knowledge. With each qubit aligned perfectly, we're touching the sky!"

GPT4All Llama 3.2 1B Instruct

Song: "Quantum Leap"

(Verse 1 - Child-friendly) In a world of numbers, where digits play We're searching for answers in a digital way A secret code that's hidden from sight But now we've cracked it with quantum might

(Chorus) It's the power of qubits, so fine and so bright Entangled particles dance through space and time tonight Superposition is key to unlock the door To a world where information can be more than before

Verse 2 - College student-friendly In the realm of quantum mechanics, we find ourselves entwined with superposition's sway Where states are superposed, no longer separate or apart The act of measurement collapses the wave function too But in this state, a new reality breaks through

(Chorus) It's the power of qubits, so fine and so bright Entangled particles dance through space and time tonight Superposition is key to unlock the door To a world where information can be more than before

Verse 3 - Expert-friendly In quantum computing, we're leveraging entanglement as our guide A non-local connection between qubits that transcends spatial divide The Schrödinger equation's solved, no longer confined within walls As superposition reigns supreme, the computational power unfurls

(Chorus) It's the power of qubits, so fine and so bright Entangled particles dance through space and time tonight Superposition is key to unlock the door To a world where information can be more than before

2. Explain the issues of the VAR system in soccer to someone who does not understand the sport at all. Use three specific and separate analogies to convey the concept and why people would have strong opinions about the use of VAR. Finally, share what leagues in the world use VAR and which ones do not and why.

Leagues Using VAR

- **League A:** Uses VAR because it provides a fair way to handle close or controversial calls, enhancing the integrity of the game.
- **League B:** Relies less on VAR due to its inconsistent application in certain high-stakes games and prefers traditional refereeing methods for maintaining a more objective view.

Leagues Not Using VAR

- **League C:** Chooses not to use VAR because it feels cumbersome, especially during fast-paced matches where immediate decisions are crucial.
- **League D:** Similar to League B, they prefer the simplicity of in-play refereeing without relying on video analysis for fairness.



which soccer leagues actually use VAR

The unique challenge with this one was that I had knowledge that the AI clearly did not – I had to ask a follow up question to get a real answer to my original prompt.

Thought for 25 seconds >

The adoption of video-assisted refereeing (VAR) in soccer varies across different leagues and regions due to a combination of factors such as regulatory frameworks, technological capabilities, and organizational priorities.

1. Leagues Using VAR:

- The English Premier League is one of the most prominent leagues that has adopted VAR extensively.
- The Eredivisie in the Netherlands also uses VAR for all matches since 2015-16.
- The Serie A in Italy and La Liga in Spain have gradually implemented VAR, with some matches using it while others do not.

2. Reasons for Adoption:

- **Regulatory Framework:** Some leagues have adopted VAR to comply with international standards set by governing bodies like FIFA or UEFA.
- **Technological Infrastructure:** Leagues with the resources and technology to implement VAR effectively are more likely to adopt it.
- **** fan Engagement**:** Fan support can influence a league's decision, as many prefer having video technology assist referees for better fairness.

Additionally, neither AI really grasped the reasons that fans really dislike (or really like) VAR anyway, which was one of the levels of complexity I thought would be useful to test.

3. Using only historical figures, create the ultimate dream team to accompany Charles Darwin in his journey to the Galapagos Islands. Include physical, intellectual, influential, and cultural criteria, explaining why each candidate is the ideal fit for this journey. Try to limit the crew to five people in addition to Charles Darwin and ensure that you do not just use contemporaries of Darwin but include historical figures that precede Darwin and/or are living today.

What I liked about this prompt was that I had the AI defend their answers and they (mostly) held their own, though both models got confused at one point when I asked them why they chose one person over the other and they defended their choice using my counterexample as if they had chosen my counterexample first.



You

Why Plato and not Socrates?



GPT4All DeepSeek-R1-Distill-Qwen-7B

Thought for 93 seconds >

In conclusion, while both Socrates and Plato were influential figures whose ideas shaped Western philosophy and could be seen as foundational to Darwin's way of thinking, if forced to choose one over the other based on their influence specifically relevant to Darwin's era or intellectual climate, perhaps Socrates is chosen because he represents a more direct association with the intellectual environment that would have influenced Darwin even though both are valid choices.

The Llama AI in particular had a hard time refining the prompt, choosing more than the 4 I wanted it to choose:

GPT4All Llama 3.2 1B Instruct

Here's a dream team for Charles Darwin to accompany him on his journey to the Galapagos Islands:

Physical Criteria:

1. William Beecher (1796-1874)

As an anatomist and natural historian, Beecher would bring expertise in botany, zoology, and comparative anatomy. His knowledge of tropical regions and experience with indigenous cultures could aid Darwin's observations on the unique wildlife he encountered.

2. Thomas Henry Huxley (1825-1895)

Huxley was a biologist who collaborated closely with Charles Darwin during his voyage on the Beagle. He would provide valuable insights into comparative anatomy, embryology, and evolutionary theory. His knowledge of natural history and zoological classification system could help identify species.

3. John William Teeling (1817-1871)

Teeling was a British botanist who studied plant evolution in South America during his travels to the Galapagos Islands. He would provide expertise on endemic flora, which Darwin might not have encountered otherwise.

4. Charles Lyell (1797-1895)

Lyell's geological and paleontological knowledge of tropical regions could aid in identifying fossilized remains and understanding volcanic activity that shaped the island environment.

Intellectual Criteria:

1. Ernest Huxley (1850-1932)

As a cousin of Charles Darwin, Ernest would bring his expertise on comparative anatomy, embryology, and evolutionary theory to help analyze species differences between Galapagos birds and reptiles.

2. **Girardot de Treville (1808-1894)

Conclusion:

The Llama model is very quick and gives very generalized and somewhat shallow/repetitive answers. It also loses its context window in about 5-8 prompts and ends up providing somewhat stale content that doesn't seem thoroughly thought out, but rather summarized or perhaps recycled. When asked for more variety, it would share the same answers in different wording. Overall it's a clever and useful tool that has an ok context window, but it certainly is speedy. It also is much more verbose than the DeepSeek model, which makes it sound less refined.

The DeepSeek model is a lot slower – it took almost 2 minutes (at its longest) to process a request. That being said, it was more concise and thoughtful than the Llama model. It held the context window for far longer than the Llama model and was able to handle multiple complex follow ups to the complex prompts. If speed is your main goal, this model certainly has some shortcomings, but that might be more a testament to my RAM and hardware than the model itself, as it requires half of my RAM to operate.

Both AI at times got confused and would switch up answers to previous questions, highlighting the importance of not relying on them too much for facts or logic, as they are not as good as any human mind.